POWER — ground star on the power-path board (left leg)

BRAIN — LOLIN S3 Mini (left middle)

SOUND, MIC, microSD

VUSB (J1: micro-USB charging socket at the heel of the left leg; D3 + C9 on the TP4056 IN pads)

C9 22µ 16V

D3 SMAJ5.0A (TVS)

R1 4.7k

R15 68k

R16 100k

J1 USB charging socket (VBUS, GND)

U1 TP4056 + DW01, R_PROG 2.4k (IN+, IN−, OUT+, OUT−, B+, B−)

Q1 AO3401A

D1 1N5822

inter. ≥ 1 A

SW1

VSYS

U9 TPS63020 3.3 V (VIN, VOUT, EN, PS, GND)

+3V3

PWR_FLAG

R2 1M (harness, B+ pad)

R3 1M

F1 PPTC 1.6A

F2 PPTC 1.6A

BT1 18650 G

BT2 18650 D

C6 0.1µ (S3)

C8 0.1µ (S3)

USB_DET

R15 / R16: charger plugged in → USB_DET ≈ 3 V (GPIO21, wake-up).

SCREENS, I2C SENSORS →

U3 Screen 1 (left middle) (GND, VCC, SCL, SDA, RES, DC, CS, BLK)

U4 Screen 2 (right middle) (GND, VCC, SCL, SDA, RES, DC, CS, BLK)

U7 DS3231 ZS-042 (right leg) (32K, SQW, SCL, SDA, VCC, GND)

U8 ADXL345 GY-291 (GND, VCC, CS, INT1, INT2, SDO, SDA, SCL)

U11 BME280 (right leg) (VCC, GND, SCL, SDA)

C1 220µ tant. CMS

C2 10µ tant. (P1)

C3 0.1µ (P1)

C10 10µ tant. (P2)

C11 0.1µ (P2)

C1 (SMD 7343 tantalum lying under the S3) on its carrier, C2 / C3 on the P1 3V3 strip, C10 / C11 on P2; plans renders/tars4_board_*.svg.

U2 LOLIN S3 Mini (EN, IO1, IO2, IO3, IO4, IO5, IO12, IO6, IO13, IO7, IO11, IO8, IO10, IO9, 3V3, IO14, IO33, IO43, IO37, IO44, IO38, IO36, IO34, IO35, IO21, IO18, IO17, IO16, GND, GND, IO15, VBUS)

R11 33

VSENSE

R12 47 (carrier)

R13 33

R14 33

R20 33 (carrier)

R19 47 (carrier)

R4 10k (P1)

R10 1k (P2)

R18 10k (P2)

U5 MAX98357A (right middle) (LRC, BCLK, DIN, GAIN, SD, GND, GND, VIN, OUT+, OUT−)

39×22×10 8Ω

LS1

C7 470µ 10V

U6 INMP441 (left middle) (SCK, WS, SD, L/R, VDD, GND)

R22 100k

Q5 AO3401A

SD_3V3

R5 10k

R6 10k

R7 10k

R8 10k

R9 10k

J2 microSD (left leg) (CLK, CMD, D0, D1, D2, D3, VCC, GND)

C4 10µ tant.

C5 0.1µ

U10 AM312 (VIN, OUT, GND)

C12 10µ tant. (P2)

R21 100 (P2)

R17 1k (P2)

VSYS SENSE — R23 in the harness at P4, R24 + Q4 on the carrier: GPIO37 low = VSYS present

R23 1M (harness, P4 end)

R24 330k (carrier)

Q4 BC547B (carrier)

U10: front hole at the top (right middle), R17 on P2; U7, U11 in the right leg (SQW not wired).

Q1 + D1 + R1: power path (mains / battery). R2 / R3 (1M, on B+: the real cell voltage, even on mains) on the power-path board; C6 (GPIO1) and C8 (GPIO21) at the S3 pins. Q5 + R22 on the power-path board: microSD cut in sleep (GPIO36 low = on).

U9: EN tied to VIN, PS/SYNC to GND (never drive it high). Its VSYS/GND pair leaves from the star; the logic GND to P1 leaves from its GND pad.

S3: V (VBUS) = the S3 USB-C only. EN = S3 reset (≠ BL_EN). 33 Ω series at the S3 (R20 on MOSI); SCK split on the carrier: R12 → screen 1, R19 → screen 2 (direct wire in the tube), 47 Ω (up to 100 if the image jitters).

BL_EN (GPIO7): backlight of both screens only (PWM possible). AMP_SD (GPIO4): R10 1k + R18 10k on P2 → amplifier SD (off at boot and in sleep). P1 / P2: stripboards under the screens. I2C chain: S3 → ADXL345 → middle tube → P2 → right tube → DS3231 + BME280 (right leg); sensor supply P1 → mic → ADXL345, P2 → right leg.

LEDs: remove every status LED fed by the battery (ZS-042, TPS63020, GY-291, microSD reader...); keep the TP4056 charge LEDs.

Silicone wire: power in 26 AWG (B+/B−, amplifier pair, 3V3 and GND to P1), signals and screen / sensor supply in 30 AWG. Tubes: 14 / 19 / 6 wires.

Connectors X1-X15 (keyed Dupont / JST) between each module and the harness, S3 plugged into a carrier: not drawn, see tars4_wiring.html. D3: unidirectional TVS (Zener symbol). A crossing without a dot is not a connection. Generated by tools/kicad_sch_gen.py (netlist checked at every generation).

LOLIN S3 Mini, TPS63020, 2 × 18650

Sheet: /
File: tars4.kicad_sch

Title: TARS4 — electronics

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